

Dataset Understanding Report

mergerd_data.csv - NHANES 2017-March 2020 Pre-Pandemic (Merged Health Survey Data)

15,560 participants x 210 columns

Prepared for EDA, Data Cleaning, Feature Engineering and Machine Learning planning

All dataset-specific claims in this report are based on direct inspection of the provided CSV file.

Domain background is drawn from official CDC/NCHS NHANES documentation and cited medical sources.

Analytical recommendations are the author's professional judgment and are labelled as such.

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1. Dataset Overview

Facts observed directly in the CSV:

- The file contains **15,560 rows and 210 columns**. The first column is an unnamed integer index (0,1,2,...) added on export - it should be dropped.
- Every row is uniquely identified by **SEQN**, a respondent sequence number. There are **0 duplicate rows and 0 duplicate SEQN values**.
- All rows have SDDSRVYR = 66, which per the official CDC codebook denotes the **NHANES 2017-March 2020 pre-pandemic** data release.
- The column names (SEQN, RIDAGEYR, BMXBMI, BPXOSY1, PAQ605, SMQ020, DIQ010, MCQ160C, LBXGH, LBDLDL, WTSAPPRP, SDMVPSU, ...) are the standard variable names of the U.S. National Health and Nutrition Examination Survey (NHANES). The file is a merge of the Demographics, Body Measures, Blood Pressure, Physical Activity, Smoking, Diabetes, Medical Conditions, Health Status, Osteoporosis, Glycohemoglobin, and standard Lipid/Glucose laboratory files.
- 5,867 of the 15,560 participants are children under 18. Many questionnaire sections are adult-only, which produces large, structurally-gated blocks of missing data.
- 1,260 participants (RIDSTATR = 1) were interviewed only and never examined at the mobile examination center (MEC); they have lab/exam measurements missing by design.

Interpretation (domain knowledge): NHANES is a continuous, nationally representative cross-sectional survey of the U.S. civilian non-institutionalized population, run by the CDC/NCHS. It combines in-home interviews, physical examinations, and laboratory tests. One row = one survey participant. Likely domain: **public health / healthcare**. Likely real-world use cases: disease-risk screening models (e.g., diabetes, hypertension, cardiovascular risk), epidemiological association studies, and health-behavior analytics.

Size and structure: 15,560 x 210 = about 3.3 million cells. Data is wide: ~210 mixed-type columns, heavily sparse, with blocks of missingness that follow NHANES skip logic (age gates, fasting subsample, exam attendance), not random dropout.

1.1 Full Column Reference Table

Missing % and Unique counts are computed from the actual file. "Role" is an analytical recommendation. Codes such as 7 (refused), 9 (don't know) and 99 exist in many categorical columns and are documented in the NHANES codebooks.

Column	Type	Description	Role / Action	Miss %	Uniq
Unnamed: 0	categorical	NHANES item - consult the NHANES codebook for exact wording	Feature	0.0	15560
SEQN	identifier	Unique respondent sequence number (participant ID)	ID - never a feature	0.0	15560
SDDSRVYR	categorical	Data release cycle - constant 66 (2017-March 2020 pre-pandemic)	Drop (constant)	0.0	1
RIDSTATR	categorical	Interview/examination status (1=interviewed+examined, 2=interviewed only)	Cohort filter, not a feature	0.0	2
RIAGENDR	categorical	Gender (1=Male, 2=Female)	Feature	0.0	2
RIDAGEYR	numerical	Age in years at screening (80 = 80+, top-coded; 574 rows contain a tiny epsilon artifact where age is missing)	Feature	0.0	81
RIDAGEMN	numerical	Age in months (only reported for children under 2 years)	Feature (age-gated; mostly missing)	93.7	25
RIDRETH1	categorical	Race/Hispanic origin, 5 categories	Feature	0.0	5
RIDRETH3	categorical	Race/Hispanic origin, 7 categories (adds non-Hispanic Asian)	Feature (redundant with RIDRETH1)	0.0	6
RIDEXMON	categorical	Six-month time period of examination (1=Nov-Apr, 2=May-Oct)	Feature (seasonality proxy)	8.1	2
DMDBORN4	categorical	Country of birth (1=USA, 2=other)	Feature	0.0	4
DMDYRUSZ	categorical	Years lived in the USA (foreign-born only)	Feature (age-gated)	80.5	6
DMDEDUC2	categorical	Education level, adults 20+ (1-5 scale)	Feature (age-gated)	40.7	7
DMDMARTZ	categorical	Marital status (1=married, 2=widowed/divorced/separated, 3=never married)	Feature (age-gated)	40.7	5
RIDEXPRG	categorical	Pregnancy status at exam (females 20-44)	Feature (age/sex-gated)	88.0	3
SIALANG	categorical	Interview language / proxy / interpreter metadata flag	Drop (administrative)	0.0	2
SIAPROXY	categorical	Interview language / proxy / interpreter metadata flag	Drop (administrative)	0.0	2
SIAINTRP	categorical	Interview language / proxy / interpreter metadata flag	Drop (administrative)	0.0	2
FIALANG	categorical	Interview language / proxy / interpreter metadata flag	Drop (administrative)	6.9	2
FIAPROXY	categorical	Interview language / proxy / interpreter metadata flag	Drop (administrative)	6.9	2
FIAINTRP	categorical	Interview language / proxy / interpreter metadata flag	Drop (administrative)	6.9	2
MIALANG	categorical	Interview language / proxy / interpreter metadata flag	Drop (administrative)	29.3	2

Column	Type	Description	Role / Action	Miss %	Uniq
MIAPROXY	categorical	Interview language / proxy / interpreter metadata flag	Drop (administrative)	29.3	2
MIINTRP	categorical	Interview language / proxy / interpreter metadata flag	Drop (administrative)	29.3	2
AIALANGA	categorical	Interview language / proxy / interpreter metadata flag	Drop (administrative)	47.1	3
WTINTPRP	numerical	Interview sample weight (survey weight, NOT a feature)	Weight - exclude from X	0.0	13537
WTMECPRP	numerical	MEC exam sample weight; 1260 epsilon artifacts (= interview-only participants)	Weight - exclude from X	0.0	12738
SDMVPSU	categorical	Masked primary sampling unit (survey design)	Design var - exclude from X	0.0	3
SDMVSTRA	categorical	Masked stratum (survey design)	Design var - exclude from X	0.0	24
INDFMPIR	numerical	Family income-to-poverty ratio (0-5; 142 epsilon artifacts = missing)	Feature	14.1	480
BMDSTATS	categorical	Body measures component status code	Drop (administrative)	8.1	4
BMXWT	numerical	Weight (kg)	Feature	9.5	1513
BMIWT	text	Body-measures comment/status code - almost entirely empty	Drop (near-empty admin)	96.2	3
BMXRECUM	numerical	Recumbent length (children)	Feature (age-gated)	90.6	512
BMIRECUM	text	Body-measures comment/status code - almost entirely empty	Drop (near-empty admin)	99.7	1
BMXHEAD	numerical	Head circumference (infants)	Feature (age-gated)	98.0	101
BMIHEAD	text	Body-measures comment/status code - almost entirely empty	Drop (near-empty admin)	100.0	0
BMXHT	numerical	Standing height (cm)	Feature	15.4	1110
BMIHT	text	Body-measures comment/status code - almost entirely empty	Drop (near-empty admin)	98.9	2
BMXBMI	numerical	Body mass index (kg/m2)	Feature	15.6	478
BMDBMIC	categorical	BMI category for children/adolescents (only <20 yrs)	Feature (age-gated)	69.5	4
BMXLEG	numerical	Upper leg length (cm)	Feature	29.4	260
BMILEG	text	Body-measures comment/status code - almost entirely empty	Drop (near-empty admin)	96.9	1
BMXARML	numerical	Upper arm length (cm)	Feature	13.3	366
BMIARML	text	Body-measures comment/status code - almost entirely empty	Drop (near-empty admin)	96.9	1
BMXARMC	numerical	Arm circumference (cm)	Feature	13.3	414
BMIARMC	text	Body-measures comment/status code - almost entirely empty	Drop (near-empty admin)	96.8	1
BMXWAIST	numerical	Waist circumference (cm)	Feature	19.2	1105
BMIWAIST	text	Body-measures comment/status code - almost entirely empty	Drop (near-empty admin)	96.0	1
BMXHIP	numerical	Hip circumference (cm)	Feature	36.6	834
BMIHIP	text	Body-measures comment/status code - almost entirely empty	Drop (near-empty admin)	97.6	1
BPAOARM	categorical	Arm used for oscillometric BP measurement (R/L); stored as raw bytes b'R'/b'L'/b" - decode before use	Feature after cleaning	25.1	3
BPAOCSZ	categorical	Cuff size code used for BP measurement	Feature (proxy for arm size)	33.3	4
BPXOSY1	numerical	Systolic BP, reading 1 (mmHg)	Feature	33.5	148
BPXODI1	numerical	Diastolic BP, reading 1 (mmHg)	Feature	33.5	99
BPXOSY2	numerical	Systolic BP, reading 2 (mmHg)	Feature (combine with readings 1-3)	33.6	147
BPXODI2	numerical	Diastolic BP, reading 2 (mmHg)	Feature (combine)	33.6	97
BPXOSY3	numerical	Systolic BP, reading 3 (mmHg)	Feature (combine)	33.9	142
BPXODI3	numerical	Diastolic BP, reading 3 (mmHg)	Feature (combine)	33.9	95
BPXOPLS1	numerical	Pulse rate, reading 1	Feature	39.1	91
BPXOPLS2	numerical	Pulse rate, reading 2	Feature (combine)	39.3	93
BPXOPLS3	numerical	Pulse rate, reading 3	Feature (combine)	39.5	92
BPQ020	categorical	Ever told by doctor you have high blood pressure (1=Yes,2=No)	Feature / target candidate (leakage risk if target is measured hypertension)	34.5	3
BPQ030	categorical	Ever told you had high BP on 2+ visits (follow-up)	Leakage risk vs hypertension target	76.9	3
BPD035	numerical	Age when first told you had high BP	Leakage risk vs hypertension target	76.9	71
BPQ040A	categorical	Currently taking BP medication (BPQ030=1 only)	Feature / leakage risk	76.9	4
BPQ050A	categorical	Now taking prescribed medicine for high BP	Feature	78.9	2
BPQ080	categorical	Ever told you had high blood cholesterol (1=Yes,2=No)	Feature	34.5	4
BPQ060	categorical	Ever had blood cholesterol checked	Feature	55.7	4
BPQ070	categorical	When was cholesterol last checked	Feature	51.5	5
BPQ090D	categorical	Currently taking cholesterol medication	Feature / leakage risk	51.5	4
BPQ100D	categorical	Taking cholesterol meds to lower cholesterol	Feature	82.3	3
PAQ605	categorical	Vigorous work activity (1=Yes,2=No)	Feature	37.7	4
PAQ610	numerical	Days per week of vigorous work activity	Feature	84.5	8
PAD615	numerical	Minutes per day of vigorous work activity	Feature	84.6	28
PAQ620	categorical	Moderate work activity	Feature	37.7	3
PAQ625	numerical	Days per week of moderate work activity	Feature	73.1	8
PAD630	numerical	Minutes per day of moderate work activity	Feature	73.2	34
PAQ635	categorical	Walk or bicycle for transportation	Feature	37.7	3
PAQ640	numerical	Days per week walking/biking to get around	Feature	85.5	8

Column	Type	Description	Role / Action	Miss %	Uniq
PAD645	numerical	Minutes per day walking/biking to get around	Feature	85.6	38
PAQ650	categorical	Vigorous recreational activity	Feature	37.7	2
PAQ655	numerical	Days per week vigorous recreation	Feature	84.4	8
PAD660	numerical	Minutes per day vigorous recreation	Feature	84.5	33
PAQ665	categorical	Moderate recreational activity	Feature	37.7	3
PAQ670	numerical	Days per week moderate recreation	Feature	74.9	8
PAD675	numerical	Minutes per day moderate recreation	Feature	75.0	36
PAD680	numerical	Minutes per day of sedentary activity (screen time); 3 epsilon artifacts	Feature	37.8	66
SMQ020	categorical	Smoked at least 100 cigarettes in lifetime (1=Yes,2=No)	Feature	37.7	4
SMD030	numerical	Age when started smoking regularly; 98 epsilon artifacts	Feature (SMQ020=1 only)	75.0	55
SMQ040	categorical	Do you now smoke cigarettes (1=every day,2=some days,3=not at all)	Feature	75.0	3
SMQ050Q	numerical	How long since quit smoking (quantity)	Feature (former smokers)	85.8	55
SMQ050U	categorical	Unit for SMQ050Q (1=days,2=months,3=years)	Feature	86.7	4
SMD057	numerical	Age last smoked regularly	Feature	85.8	35
SMQ078	categorical	How soon after waking do you smoke	Feature (smokers)	91.7	9
SMD641	numerical	Number of cigarettes smoked per day now; 51 epsilon artifacts	Feature (smokers)	88.8	30
SMD650	numerical	Number of cigarettes per day during past 30 days	Feature	89.2	32
SMD100FL	categorical	Cigarette filter type; 30 epsilon artifacts	Feature	89.8	3
SMD100MN	categorical	Menthol cigarette indicator; 807 epsilon artifacts	Feature	89.8	3
SMQ670	categorical	Used tobacco/nicotine last 30 days	Feature	89.0	2
SMQ621	categorical	Ever used e-cigarettes	Feature	91.2	9
SMD630	categorical	Used e-cigarettes in past 30 days	Feature	99.6	10
SMAQUX2	categorical	Smoking questionnaire status code	Drop (administrative)	28.4	2
HSQ590	categorical	Self-rated general health (1=poor ... 5=excellent)	Feature / target candidate	43.2	3
DIQ010	categorical	Doctor told you that you have diabetes (1=Yes, 2=No, 3=Borderline, 9=Refused) - PRIMARY TARGET CANDIDATE	TARGET (binary classification after cleaning)	3.7	4
DID040	numerical	Age when first told you had diabetes (follow-up, DIQ010=1) - LEAKAGE if DIQ010 is the target	LEAKAGE - drop	90.7	82
DIQ160	categorical	Ever told you are at risk for diabetes / prediabetes (questionnaire item)	Feature (near-target; handle with care)	38.8	3
DIQ180	categorical	Ever had blood tested for high blood sugar	Feature	37.0	4
DIQ050	categorical	Taking insulin now (DIQ010=1 only) - LEAKAGE if DIQ010 is the target	LEAKAGE - drop	90.7	3
DID060	numerical	How long taking insulin	LEAKAGE - drop	97.3	46
DIQ060U	categorical	Insulin unit per day and how long taking insulin	LEAKAGE - drop	97.3	2
DIQ070	categorical	Taking diabetic pills (DIQ010=1 only) - LEAKAGE if DIQ010 is the target	LEAKAGE - drop	82.8	3
DIQ230	categorical	Seen a diabetes specialist in past year (DIQ010=1) - LEAKAGE	LEAKAGE - drop	90.7	6
DIQ240	categorical	Seen any other provider for diabetes (DIQ010=1) - LEAKAGE	LEAKAGE - drop	90.7	3
DID250	categorical	Last time had A1C test; 19 epsilon artifacts	LEAKAGE - drop	92.9	20
DID260	numerical	A1C level last time; 300 epsilon artifacts	LEAKAGE - drop	90.7	18
DIQ260U	categorical	Unit for DID260	LEAKAGE - drop	92.7	4
DIQ275	categorical	SBP control goal last time; questionnaire item	LEAKAGE - drop	90.7	3
DIQ280	categorical	Last time LDL cholesterol checked	LEAKAGE - drop	92.2	75
DIQ291	categorical	LDL cholesterol goal	LEAKAGE - drop	92.2	8
DIQ300S	numerical	SBP last time checked by doctor	LEAKAGE - drop	90.8	99
DIQ300D	numerical	DBP last time checked by doctor	LEAKAGE - drop	90.8	78
DID310S	numerical	SBP control goal	LEAKAGE - drop	90.8	46
DID310D	numerical	DBP control goal	LEAKAGE - drop	90.8	45
DID320	categorical	Last time feet checked for sores; 381 epsilon artifacts	LEAKAGE - drop	90.8	126
DID330	categorical	Last time eyes examined	LEAKAGE - drop	91.3	45
DID341	categorical	Diabetes affected eyes; 260 epsilon artifacts	LEAKAGE - drop	90.8	26
DID350	categorical	Seen a foot specialist in past year; 300 epsilon artifacts	LEAKAGE - drop	90.8	16
DIQ350U	categorical	Unit for time since last foot specialist visit	LEAKAGE - drop	92.5	4
DIQ360	numerical	Times checked for glycemic control per year	LEAKAGE - drop	90.8	6
DIQ080	categorical	Ever told you have diabetes only during pregnancy (females)	Feature (gestational diabetes history)	90.8	3
MCQ010	categorical	Ever told you have asthma	Feature	3.7	3
MCQ025	numerical	Age when first had asthma	Feature	85.1	80
MCQ035	categorical	Still have asthma	Feature	85.1	4
MCQ040	categorical	Had asthma attack in past year	Feature	90.9	4
MCQ050	categorical	ER visit for asthma in past year	Feature	90.9	2
AGQ030	categorical	Allergic to any foods	Feature	57.5	3
MCQ053	categorical	Ever told you have hay fever/allergic rhinitis	Feature	3.7	4

Column	Type	Description	Role / Action	Miss %	Uniq
MCQ080	categorical	Ever told you are overweight (self-report)	Feature	34.5	3
MCQ092	categorical	COPD diagnosis follow-up item	Feature	15.1	4
MCD093	categorical	COPD follow-up detail	Feature	92.8	4
MCQ149	categorical	Mouth/teeth condition item	Feature	95.6	2
MCQ151	categorical	Lost all permanent teeth item	Feature	99.4	5
RHD018	categorical	Heart condition follow-up item	Feature	99.4	29
MCQ160A	categorical	Ever told you have arthritis	Feature	40.7	3
MCQ195	categorical	Type of arthritis	Feature	81.9	6
MCQ160B	categorical	Ever told you have congestive heart failure	Feature / target candidate	40.7	3
MCD180B	numerical	Age first told you had CHF (follow-up) - LEAKAGE if MCQ160B is target	LEAKAGE - drop	97.7	62
MCQ160C	categorical	Ever told you have coronary heart disease	Feature / target candidate	40.7	3
MCD180C	numerical	Age first told you had CHD (follow-up) - LEAKAGE	LEAKAGE - drop	97.3	55
MCQ160D	categorical	Ever told you have angina/angina pectoris	Feature	40.7	3
MCD180D	numerical	Age first told you had angina	LEAKAGE - drop	98.5	58
MCQ160E	categorical	Ever told you have a heart attack (myocardial infarction)	Feature / target candidate	40.7	3
MCD180E	numerical	Age first told you had heart attack	LEAKAGE - drop	97.2	59
MCQ160F	categorical	Ever told you had a stroke	Feature / target candidate	40.7	3
MCD180F	numerical	Age first told you had stroke	LEAKAGE - drop	96.9	65
MCQ160M	categorical	Ever told you have thyroid problem	Feature	40.7	4
MCQ170M	categorical	Still have thyroid problem	Feature	93.1	4
MCD180M	numerical	Age first told you had thyroid problem	LEAKAGE - drop	93.1	66
MCQ160P	categorical	Ever told you have any kind of liver condition	Feature / target candidate	40.7	3
MCQ160L	categorical	Ever told you have any kind of liver condition (long form)	Feature / target candidate	40.7	3
MCQ170L	categorical	Still have liver condition	Feature	97.0	3
MCD180L	numerical	Age first told you had liver condition	LEAKAGE - drop	97.0	66
MCQ500	categorical	Blood transfusion follow-up item	Feature	87.1	3
MCQ510A	categorical	Hepatitis condition item A (near-empty)	Feature (96%+ missing)	98.5	2
MCQ510B	categorical	Hepatitis condition item B (10 non-null)	Drop (near-empty)	99.9	1
MCQ510C	categorical	Hepatitis condition item C (34 non-null)	Drop (near-empty)	99.8	1
MCQ510D	categorical	Hepatitis condition item D (81 non-null)	Drop (near-empty)	99.5	1
MCQ510E	categorical	Hepatitis condition item E (20 non-null)	Drop (near-empty)	99.9	1
MCQ510F	categorical	Hepatitis condition item F (118 non-null)	Drop (near-empty)	99.2	1
MCQ520	categorical	Jaundice follow-up item	Feature	40.7	3
MCQ530	categorical	Hepatitis B/C screening item	Feature	86.9	4
MCQ540	categorical	Ever tested for hepatitis B	Feature	86.9	3
MCQ550	categorical	Ever tested for hepatitis C	Feature	40.7	3
MCQ560	categorical	Ever told you have gallstones	Feature	40.7	3
MCQ570	numerical	Age first told you had gallstones	Feature	93.4	70
MCQ220	categorical	Ever told you had cancer/malignancy	Feature / target candidate	40.7	3
MCQ230A	numerical	Age at first cancer diagnosis (first type)	Feature / leakage risk vs cancer target	93.5	31
MCQ230B	numerical	Age at cancer diagnosis (second type)	Feature (99% missing)	99.3	24
MCQ230C	numerical	Age at cancer diagnosis (third type)	Drop (99.9% missing)	99.9	10
MCQ230D	numerical	Age at cancer diagnosis (fourth type)	Drop (4 non-null)	100.0	1
MCQ300B	categorical	Family history - close relative had heart attack	Feature	15.1	3
MCQ300C	categorical	Family history - close relative had diabetes	Feature	40.7	4
MCQ300A	categorical	Family history - close relative had stroke	Feature	40.7	4
MCQ366A	categorical	Medical conditions questionnaire item (see codebook)	Feature	34.5	4
MCQ366B	categorical	Medical conditions questionnaire item (see codebook)	Feature	34.5	3
MCQ366C	categorical	Medical conditions questionnaire item (see codebook)	Feature	34.5	3
MCQ366D	categorical	Medical conditions questionnaire item (see codebook)	Feature	34.5	3
MCQ371A	categorical	Medical conditions questionnaire item (see codebook)	Feature	34.5	3
MCQ371B	categorical	Medical conditions questionnaire item (see codebook)	Feature	34.5	3
MCQ371C	categorical	Medical conditions questionnaire item (see codebook)	Feature	34.5	3
MCQ371D	categorical	Medical conditions questionnaire item (see codebook)	Feature	34.5	4
OSQ230	categorical	Ever told you have osteoporosis	Feature / target candidate	58.7	3
LBXGH	numerical	Glycohemoglobin (HbA1c) - % of glycated hemoglobin; SECONDARY TARGET (regression)	TARGET (regression) / Feature for DIQ010	37.4	109
WTSAPFRP	numerical	Fasting subsample weight; 614 epsilon artifacts	Weight - exclude from X	67.3	4351
LBXGLU	numerical	Fasting plasma glucose (mg/dL) - fasting subsample only (~69.5% missing)	Feature / target candidate	69.5	240

Column	Type	Description	Role / Action	Miss %	Uniq
LBDGLUSI	numerical	Fasting glucose in SI units (mmol/L) - exact unit duplicate of LBXGLU	Drop (redundant)	69.5	209
LBDHDD	numerical	HDL cholesterol (mg/dL)	Feature	30.4	123
LBDHDDSI	numerical	HDL cholesterol, SI units (mmol/L) - unit duplicate of LBDHDD	Drop (redundant)	30.4	123
LBXTC	numerical	Total cholesterol (mg/dL)	Feature	30.4	272
LBDTCSI	numerical	Total cholesterol, SI units - unit duplicate of LBXTC	Drop (redundant)	30.4	272
WTSAFPRP_P_TRIGLY	numerical	Exact duplicate column of WTSAFPRP	Drop (duplicate)	67.3	4351
LBXTR	numerical	Triglycerides (mg/dL) - fasting subsample only	Feature	70.1	354
LBDTRSI	numerical	Triglycerides, SI units - unit duplicate of LBXTR	Drop (redundant)	70.1	354
LBDLDL	numerical	LDL cholesterol (mg/dL) - fasting subsample only	Feature	70.3	213
LBDLDLSI	numerical	LDL cholesterol, SI units - unit duplicate of LBDLDL	Drop (redundant)	70.3	213
LBDLDLM	numerical	LDL (mg/dL), Martin/Hopkins method (fasting)	Feature (redundant with LBDLDL)	70.3	212
LBDLDMSI	numerical	LDL Martin method, SI units	Drop (redundant)	70.3	212
LBDLDLN	numerical	LDL (mg/dL), NCEP method (fasting)	Feature (redundant with LBDLDL)	70.2	215
LBDLDNSI	numerical	LDL NCEP method, SI units	Drop (redundant)	70.2	215

2. Target Variable

Observed in the CSV (DIQ010): 1 = Yes: 1,445; 2 = No: 13,249; 3 = Borderline: 284; 9 = Refused: 8; missing: 574. After excluding codes 3/9 and missing, the positive rate is $1,445 / 14,694 = 9.8\%$ - a mildly imbalanced binary target.

Primary recommendation - binary classification: predict doctor-diagnosed diabetes, $y = (DIQ010 == 1)$, dropping rows with codes 3, 9 or NaN (~9.0% of rows). Suitable: 14,694 labeled rows of which 9.8% positive.

Secondary recommendation - regression: LBXGH (glycated hemoglobin / HbA1c, %). Observed in CSV: $n = 9,737$ non-null; mean 5.77, SD 1.07, median 5.5, range 2.8-16.2. Right-skewed continuous outcome; thresholds of clinical interest are 5.7% (prediabetes) and 6.5% (diabetes).

Other usable targets (observed class counts, Yes/No/Don't-know):

Candidate target	Class counts (Yes/No)	Type	Positive rate	Verdict
DIQ010 - Diabetes (self-report)	1,445 / 13,249	Binary	9.8%	PRIMARY
LBXGH - HbA1c (%)	continuous	Regression	$n=9,737$	SECONDARY
LBXGLU - Fasting glucose (mg/dL)	continuous	Regression	fasting subsample	alternative
HSQ590 - Self-rated poor health	3,376 / 5,203	Binary-ish (5-level ordinal)	39.4% poor-ish	alternative
MCQ160B - Congestive heart failure	361 / 8,848	Binary	3.9%	hard (very rare)
MCQ160C - Coronary heart disease	423 / 8,781	Binary	4.6%	hard (very rare)
MCQ160E - Heart attack	see codebook	Binary	rare	hard (very rare)
MCQ160F - Stroke	487 / 8,729	Binary	5.3%	hard (very rare)
BPQ020 - Ever told high blood pressure	3,597 / 6,586	Binary	35.3%	alternative
MCQ220 - Ever told cancer	see codebook	Binary	rare	hard (very rare)

Notes: The MCQ160x conditions are asked only of adults 20+, and each positive class has fewer than ~500 cases - workable only with strong regularization and appropriate metrics (PR-AUC, not accuracy). All diagnosis-style targets are self-reported ("told by a doctor"), so they reflect both disease and healthcare access. There is no single mandated target; the table above gives evidence-based options.

3. Domain-Specific Knowledge

Domain (determined from the data): public health / healthcare - specifically the NHANES survey. Key concepts needed to work with this dataset:

- NHANES structure:** continuous cross-sectional survey (CDC/NCHS); interviews at home + exams and lab tests in Mobile Examination Centers (MECs). Variables use fixed prefixes: RI*/DM* demographics, BMX* body measures, BPX* measured BP, BPQ* BP questionnaire, PAQ*/PAD* physical activity, SMQ*/SMD* smoking, DIQ*/DID* diabetes, MCQ*/MCD* medical conditions, HSQ* health status, OSQ* osteoporosis, LBX*/LBD* laboratory results, WT* sample weights, SDMV* masked survey design.
- Complex survey design:** NHANES uses a multistage, clustered, unequal-probability design with oversampling of Hispanics, non-Hispanic Blacks, non-Hispanic Asians, low-income persons, and adults 80+. SDMVPSU, SDMVSTRA and the WT* weights encode this. Any estimate meant to generalize to the U.S. population must use survey-weighted methods; an unweighted ML model learns the (biased) sample, not the population.
- Skip logic / structural missingness:** most questions are asked only of eligible participants (adults 18/20+, current smokers, diabetics, fasting subsample). Missingness here is informative by design, not at random.
- Coding conventions:** 1/2 = Yes/No, 7 = refused, 9 = don't know, 99/999/9999 = refused/don't know/missing, 6666/7777 = special fill codes in some BP variables. Adults 80+ are top-coded to age 80 (RIDAGEYR).

3.1 Conditions present in this dataset

Condition	Definition	Common risk factors	Typical symptoms	How identified	Notes for this dataset
Type 2 diabetes / prediabetes	Chronic hyperglycemia from insulin resistance (and, in T1D, autoimmune insulin deficiency). Prediabetes = elevated glucose/A1c not yet diagnostic.	Obesity, age, family history (MCQ300C), physical inactivity, race/ethnicity, gestational diabetes (DIQ080), hypertension, dyslipidemia.	Often asymptomatic; thirst, polyuria, fatigue.	Fasting glucose ≥ 126 mg/dL, A1c $\geq 6.5\%$, or 2h OGTT ≥ 200 ; prediabetes A1c 5.7-6.4%.	Main target here (DIQ010 / LBXGH). Distinct from borderline (code 3 in DIQ010) and gestational diabetes.

Condition	Definition	Common risk factors	Typical symptoms	How identified	Notes for this dataset
Hypertension	Persistently elevated arterial BP (>=130/80 or >=140/90 mmHg depending on guideline).	Age, obesity, high sodium intake, smoking, physical inactivity, genetics.	Usually silent ("silent killer"); severe cases cause headache, dyspnea.	Repeated office BP measurements (BPXOSY/DI) or self-report (BPQ020) plus medication history.	BPXOSY1-3/DI1-3 are the raw measurements; BPQ020 is self-report.
Congestive heart failure (MCQ160B)	Heart cannot pump/fill adequately; congestion in lungs/system.	Prior MI, hypertension, diabetes, valvular disease.	Dyspnea, orthopnea, edema, fatigue.	Clinical (echo, BNP); in survey data = self-report.	Very rare here (3.9% positives) - hard ML target.
Coronary heart disease / angina (MCQ160C/D, MCQ160E)	Atherosclerosis of coronary arteries; angina = chest pain from ischemia; MI = heart attack.	Smoking (SMQ*), LDL cholesterol (LBDLDL), hypertension, diabetes, obesity, family history (MCQ300B).	Chest pain, shortness of breath; MI may be painless.	ECG, troponin, angiography; survey = self-report.	Cardiovascular cluster - expect correlations among MCQ160B/C/D/E/F.
Stroke (MCQ160F)	Acute cerebrovascular event (ischemic or hemorrhagic).	Hypertension, diabetes, smoking, atrial fibrillation, high cholesterol.	Sudden focal deficits: weakness, aphasia, facial droop.	Neuroimaging; survey = self-report.	Family history available via MCQ300A.
Obesity / overweight	Excess adiposity; adult BMI >=30 (obese), 25-29.9 (overweight); child BMI uses age/sex percentiles (BMDBMIC).	Energy imbalance, sedentary behavior (PAD680), diet, genetics.	Usually asymptomatic; increases all risks above.	BMI from measured height/weight (BMXBMI), waist circumference (BMXWAIST).	Central variable linking lifestyle labs and outcomes.
Dyslipidemia	Abnormal lipids: high LDL (LBDLDL) or triglycerides (LBXTR), low HDL (LBDHDD).	Diet, obesity, diabetes, hypothyroidism, genetics.	Asymptomatic until atherosclerosis develops.	Lipid panel from fasting blood draw.	Total cholesterol (LBXTC) and LDL exist in SI-unit duplicates - drop duplicates.
Osteoporosis (OSQ230)	Reduced bone mineral density, fracture risk.	Age, female sex, menopause, low BMI, steroids.	Often silent until fracture.	DXA scan; survey = self-report.	Adults-only, low prevalence.
Asthma / COPD / hay fever (MCQ010, MCQ053, MCQ092)	Chronic inflammatory airway diseases; COPD = usually smoking-related airflow limitation.	Smoking (SMQ*), allergens, air pollution, occupational exposure.	Wheeze, cough, dyspnea.	Spirometry; survey = self-report plus attack/ER follow-ups (MCQ040/050).	
Liver conditions / gallstones / hepatitis screening (MCQ160P/L, MCQ520-570)	Group of hepatic diagnoses: fatty liver, hepatitis B/C history, gallstones.	Obesity, alcohol, viral hepatitis, cholesterol.	Often asymptomatic; jaundice, RUQ pain when symptomatic.	LFTs, imaging, serology; survey = self-report + screening items.	Hepatitis sub-items (MCQ510A-F) are near-empty.

Terminology: **HbA1c / glycohemoglobin (LBXGH)** = percent of hemoglobin coated with glucose, reflecting average blood sugar over ~3 months; normal <5.7%, prediabetes 5.7-6.4%, diabetes >=6.5%. **MEC** = Mobile Examination Center.

PSU/Strata = primary sampling units / strata of the survey design. **SEQN** = respondent ID.

4. Important Relationships to Investigate in EDA

These are hypotheses about associations to verify during EDA - not causal claims. All can be computed from the actual columns.

Relationship	Variables	Expected pattern (domain knowledge)	EDA decision it informs
RIDAGEYR vs BPXOSY* (systolic)	Age vs measured systolic BP (and pulse BPXOPLS*)	Blood pressure rises with age; check non-linearity (flattening in treated elderly).	Decide on age transformations/splines; verify BP-outlier cleaning matters more at extremes of age.
BMXBMI / BMXWAIST vs LBXGH, LBXGLU, LBDLDL	Adiposity vs glycemia and lipids	Obesity is the dominant modifiable risk factor for prediabetes/diabetes and dyslipidemia.	Which anthropometry (BMI vs waist vs waist-hip) carries most signal; consider waist-to-hip ratio feature.
BMXBMI vs BPXODI*	Adiposity vs diastolic BP	Central obesity correlates with diastolic elevation in younger adults.	Age-stratify (relationship differs <40 vs >60).
SMQ040 / SMD641 vs MCQ092 (COPD), BP, LBXGH	Smoking vs lung disease, BP, A1c	Smoking worsens COPD risk; associated with higher BP and A1c.	Build a composite smoking-intensity feature; check rare COPD class.
PAQ605-670 / PAD680 (activity & sedentary time) vs BMXBMI, LBXGH	Physical activity vs obesity and glycemia	Activity shows protective association with BMI and A1c.	Aggregate activity minutes; watch zero-inflation (many inactive participants).
INDFMPIR (income-to-poverty) vs LBXGH, BP, MCQ160*	Socioeconomic status vs outcomes	Lower income associated with worse control and higher comorbidity burden.	Include INDFMPIR in model; consider interactions with race/education.
RIDRETH3 vs LBXGH, BMXBMI, BP	Race/Hispanic origin vs outcomes	Documented disparities in diabetes/obesity prevalence by race/ethnicity.	Fairness-aware evaluation; check whether model performance differs across groups.
LBXTC vs LBDHDD vs LBDLDL vs LBXTR	Lipids among themselves	Total cholesterol decomposes into HDL + LDL + VLDL-ish fraction; triglycerides correlate with the remnant fraction.	Multicollinearity check; dropping SI duplicates is not enough - consider keeping one lipid summary.
BPXOSY1/2/3 within-participant consistency	Three BP readings per person	Readings correlate strongly but have measurement noise; averages are more stable.	Combine into mean feature; quantify within-person SD as a data-quality flag.
DIQ010 vs LBXGH (target vs lab)	Diagnosis vs measured A1c	Most diagnosed diabetics have elevated A1c, but some controlled diabetics look normal and some undiagnosed people have high A1c (gap between lab and self-report).	Understand label noise; motivates using LBXGH as regression target or as a feature for screening models.
LBXGH vs LBXGLU	A1c vs fasting glucose	Same underlying dysglycemia, different windows (3 months vs point-in-time).	Redundancy check; LBXGLU only exists for the fasting subsample.

Redundancy groups found in the CSV: (1) SI-unit duplicates LBDxxxSI of LBXxx/LBDxxx; (2) three BP readings + three pulse readings per participant; (3) RIDRETH1 vs RIDRETH3 (overlapping race codings); (4) LBDLDL/LBDLDLM/LBDLDLN (three LDL calculation methods); (5) WTSAFPRP duplicated as WTSAFPRP_P_TRIGLY; (6) MCQ160P vs MCQ160L (overlapping liver-condition items).

5. Data Quality Analysis (from direct inspection of the CSV)

Issue	What was found in the CSV (facts)	Why it matters for EDA/ML	Recommended handling	Severity
Artifact values 5.397605346934028e-79	Found in 15 numeric columns (e.g., WTMECPRP 1,260 rows, SMD030 98, PAD680 3, INDFMPIR 142, RIDAGEYR 574, DID250 19, LBXGLU/LBDLDL etc. in the fasting subsample). This is the float representation of SAS missing-value flags from the original XPT transport files; it is numerically ~0 but not NaN.	If treated as real values, these become fake zeros/babies: e.g., RIDAGEYR=5e-79 creates phantom 0-year-olds; LBXGLU=5e-79 creates a fake glucose of zero. Descriptive stats, correlations and any distance-based model are corrupted.	Immediately replace with NaN: <code>df.replace(5.397605346934028e-79, np.nan)</code> . Then treat as ordinary missing data.	Critical - fix before anything else.
Bytes-object column BPAOARM	Column has dtype object with values like b'R', b'L', b'' (a Python bytes literal written into the CSV, including an empty bytes value).	Cannot be used directly; many encoders will fail or treat b'R' and b'L' as distinct opaque strings, and b'' will look like a third category.	Decode: strip b" quotes to get R/L/empty, map empty to NaN, then encode R/L as binary (or drop - low information).	High - breaks most pipelines silently.
Unnamed: 0 index column	First column is a 0..15559 row index added on export.	Leakage-free but pure noise; some feature selectors may rank it spuriously.	Drop on load.	Trivial.
Constant / near-constant columns	SDDSRVYR is constant (66). 14 columns have <=1 unique value. MCQ510B-F have 10-118 non-null values (96-99.9% missing). BMIxxx comment columns (BMIWT, BMIRECUM, ...) are ~100% missing. SMAQUEX2-type status codes are administrative.	Carry no signal; waste df of tree models; near-empty columns break one-hot and median-imputation assumptions.	Drop columns with >=95-99% missing and all single-valued/admin columns. Decide the threshold after inspecting each group.	High.
64 columns with >90% missing	Concentrated in: adult-only questionnaires (children's rows are structurally empty), diabetic-only follow-ups, fasting-subsample labs, and comment/status fields.	Naive complete-case analysis would shrink n from 15,560 to a few hundred; naive mean-imputation would fabricate adult values for children.	Separate "structural" missingness (skip logic) from item non-response. Restrict the modeling cohort first (e.g., adults 18+), then impute within-pipeline.	Critical for sample size planning.
Children vs adults mixed in one table	5,867 rows are under 18. Education, marital status, most MCQ diagnoses, smoking, and many PAQ items are adult-only.	A single model mixing both groups learns age-gated missingness patterns instead of biology.	Decide the cohort up front: all-ages (age as feature), adults 18+ (recommended for DIQ010), or children (separate analysis).	Critical design decision.
Invalid / sentinel codes inside numeric fields	BPXOSY1-3 / BPXOD1-3 contain fill codes such as 9999 and 6666 (seen in the raw rows); other NHANES fields use 7777. BMXBMI max is 92.3 (extreme but biologically possible; verify unit plausibility per row).	Sentinels are 10-100x larger than real values; a single 9999 systolic reading would dominate any scaler or split.	Map 6666/7777/9999 and friends to NaN per the NHANES codebook before any statistics.	High.
Label codes inside target	DIQ010 has codes 3 (borderline, 284 rows) and 9 (refused, 8 rows) besides 1/2.	A model trained with 3 as a third class or as "negative" changes the question being answered; refused rows are unusable.	Define y strictly as (DIQ010==1); drop 3/9/NaN from training rows (document the decision).	High.
Duplicate information columns	WTSAPFRP_P_TRIGLY is an exact duplicate of WTSAPFRP; LBDxxxSI are unit conversions of LBXxx/LBDxxx; three LDL methods exist.	Perfect collinearity can break linear models and inflate feature importance.	Drop SI duplicates (or keep SI and drop mg/dL - pick one system), pick one LDL method.	Medium.
No true duplicate rows / IDs	0 duplicate rows; SEQN unique. BPAOCSZ + BP readings allow near-duplicate measurement rows only within a participant (by design).	None - but verify again after merges on your side.	Keep SEQN as index; do not use it as a feature.	Informational.
Wrong-looking data types	Many coded categoricals are float64 (1.0/2.0/7.0/9.0); BPAOARM is object-bytes; the index column is int.	Floats are fine for trees but wasteful; bytes object is not.	Downcast categoricals to Int64/category after sentinel cleaning; fix BPAOARM as above.	Medium.

Outliers to expect after sentinel cleaning: extreme-but-real values in BMXWT/BMXBMI (BMI up to 92.3), very high BP readings (verify against 9999 fills), and right-skewed labs (triglycerides, glucose). Use robust statistics (median, IQR) and visual checks rather than automatic 3-sigma deletion.

6. Data Leakage Analysis

Leakage is the most dangerous issue in this dataset because the diabetes questionnaire block contains many variables that are only asked AFTER a diabetes diagnosis.

Columns at risk	What they are	Leakage mechanism and required action
DID040, DIQ050, DIQ060U, DID060, DIQ070, DIQ230, DIQ240, DID250, DID260, DIQ260U, DIQ275, DIQ280, DIQ291, DIQ300S/D, DID310S/D, DID320, DID330, DID341, DID350, DIQ350U, DIQ360	All are follow-up questions asked ONLY to participants who answered DIQ010=1 (diabetics): age at diagnosis, insulin use, pill use, specialist visits, self-monitoring habits, complication checks.	Perfect/near-perfect proxies for the target. A model would learn "if any diabetes-management variable is non-null, predict diabetic" and score ~100% on the trivial pattern while knowing nothing about risk. Must be excluded from X when DIQ010 is the target.
DIQ160 ("told you are at risk for diabetes"), DIQ180	Near-target awareness items.	Highly correlated with diagnosis and care-seeking; likely safe to exclude to avoid proxy leakage and label noise.
MCD180B/C/D/E/F/M/L (age at first diagnosis) and MCQ170x ("still have")	Follow-ups to the MCQ160x diagnosis items.	Leakage if the corresponding MCQ160x is a target.
BPD035, BPQ030 (vs hypertension target); MCQ230A (vs cancer target)	Diagnosis-derived follow-ups.	Same pattern: exclude whenever the parent diagnosis is the target.
SEQN and the Unnamed: 0 index	Unique identifiers.	No predictive meaning; tree models can memorize IDs. Exclude.

Columns at risk	What they are	Leakage mechanism and required action
WTINTPRP / WTMECPRP / WTSAPRP survey weights	Weights encode selection probability, oversampling and non-response adjustment.	They are legitimate only for population-weighted statistics; as ML features they inject the sampling design into predictions and do not exist for a new patient. Exclude from X.
Post-outcome measurement caveat	Labs (LBXGH, LBDLDL) and medications (BPQ050A, BPQ090D) are measured/asked at the same visit as the diagnosis questions.	Not "future" data, but note the clinical direction: for a screening scenario, "currently taking insulin/pills" is post-diagnosis. Also LBXGH is essentially a cheaper re-measurement of the diabetes label - excellent for screening models, circular if the goal is "predict diagnosis without any lab". Define the use case first.
Duplicate/related observations	One row per participant; no repeated measures; no groups beyond household (household ID not present).	Standard random splitting is safe from the duplicate-observation angle.

7. Train/Test Split Strategy

Facts that drive the choice: cross-sectional survey; one row per participant; no time column; no natural groups (household IDs are not in this extract); two possible cohorts (all ages vs adults); a fasting subsample for glucose/lipids.

Recommended: stratified random split (no shuffling concerns), after cohort restriction. Stratify on the binary target $y = (\text{DIQ010}==1)$. Rationale: (a) there is no temporal ordering to respect - a time-based split is impossible and unnecessary; (b) there is no grouping unit to leak across - a group split is unnecessary; (c) the target is imbalanced (9.8%), so stratification prevents an unlucky all-negative test fold. Use 80/20 (train/test) and carve validation from train, or use 5-fold stratified cross-validation for tuning with a held-out test set.

- **Split AFTER cohort restriction and cleaning:** first drop the Unnamed index, fix the epsilon artifact and sentinels, restrict to your analysis population (e.g., adults 18+ for DIQ010), and only then split. Deciding the cohort after seeing test metrics is a subtle form of leakage.
- **Fit all preprocessing on the training fold only:** sentinel-to-NaN mapping is a fixed codebook rule (safe anywhere), but imputation values, category levels, scaling parameters and any outlier thresholds must be learned from train and applied to test.
- **Put everything tunable inside a Pipeline:** ColumnTransformer + SimpleImputer + StandardScaler + estimator. This guarantees the imputer/scaler never sees test rows.
- **Feature engineering that uses only a single row and fixed rules** (BP averages, waist-to-hip ratio, smoking pack-years) may be computed before the split safely because it involves no fitting; still, wrap it in the pipeline for reproducibility. Anything that estimates from the data (rare-category grouping, target encoding) must be fit post-split.
- **Fasting-subsample complication:** if you include LBXGLU/LBXTR/LBDLDL, you implicitly restrict to the fasting cohort ($n \sim 4,700\text{--}9,700$ depending on variable) with its own weights. Preferred options: (1) build the model on HbA1c (LBXGH, measured for all examined) and skip fasting labs, or (2) build a separate fasting-subsample model and evaluate within it. Do not mix within one model without a missingness indicator.
- **Class imbalance handling (fit inside CV only):** if using class weights or SMOTE, apply them within each training fold, never on the full dataset before splitting.

8. EDA Roadmap (based only on columns that exist)

8.1 Univariate Analysis

Visualization	Columns	What to look for	Decision informed
Histogram + KDE (with missing bars)	RIDAGEYR (after epsilon fix)	Age distribution is bumpy by design (80+ top-coded); check the child/adult mix.	Decide cohort; consider age bands.
Histogram	BMXBMI	Skew; extreme obesity tail; verify 92.3 max is real.	Outlier policy; log-transform candidates.
Histogram	LBXGH	Bimodal-ish tail above 6.5%?	Confirms target regression difficulty; motivates classification alternative.
Histogram + boxplot	LBXGLU, LBDLDL, LBXTR (fasting)	Right skew is expected for glucose/triglycerides.	Log-transform; remember subsample restriction.
Count plots	RIAGENDR, RIDRETH3, DMDEDUC2, DMDMARTZ, DIQ010, BPQ020, SMQ040	Composition of the sample; oversampling effects.	Encoding strategy; fairness slices.
Missingness matrix / heatmap	All columns, grouped by prefix	Visualize the block structure (adult-only, diabetic-only, fasting-only columns).	Justifies cohort restriction; guides imputation.
Bar of non-null % per column	All columns	Finds the 64 >90%-missing columns quickly.	Column-drop list.

8.2 Bivariate Analysis

Visualization	Columns	What to look for	Decision informed
Scatter + LOESS	RIDAGEYR vs BPXOSY*	Curvature; treatment dips.	Age feature engineering.
Scatter / hexbin	BMXBMI vs LBXGH; BMXWAIST vs LBDLDL	Monotonic, roughly linear trend with heteroscedasticity.	Confirms central role of adiposity features.
Boxplots	DIQ010 vs LBXGH, BMXBMI, BPXOSY1	Separation between classes; overlaps show label noise.	Feature selection; threshold sanity checks.
Boxplots / violin	RIDRETH3 vs LBXGH, BMXBMI	Group differences in levels and spread.	Fairness evaluation plan.
Correlation heatmap	BMXWT, BMXHT, BMXBMI, BMXWAIST, BMXHIP, LBDHDD, LBXTC, LBDLDL, LBXTR, LBXGH, BPXOSY1, BPXODI1	Multicollinearity clusters (anthropometry; lipids; BP readings).	Which redundant columns to drop.

Visualization	Columns	What to look for	Decision informed
Cross-tab with chi-square	RIAGENDR x SMQ040; RIDRETH3 x DIQ010	Association strength vs class balance.	Categorical encoding and sampling weights awareness.
Pairwise within-person	BPXOSY1 vs BPXOSY2 vs BPXOSY3 (and DI)	Reading reliability; average vs last reading.	BP aggregation rule.

8.3 Multivariate Analysis

- Pairplot / corner plot of LBXGH, LBXGLU, BMXBMI, BMXWAIST, LBDLDL, BPXOSY1 colored by DIQ010 - reveals joint structure (e.g., the lean-but-hyperglycemic cluster).
- Heatmap of correlations within the three BP readings and three pulse readings - quantifies how much averaging helps.
- BMI-category x Age-band x Gender grid of mean LBXGH (small-multiples bar chart) - shows where diabetes risk concentrates and whether interactions exist.
- PCA (after scaling) on the cleaned continuous block - check whether 2-3 components separate diabetics; useful only as an exploratory view, not necessarily for modeling.
- Grouped missingness by RIDSTATR (interviewed-only vs examined) - confirm which variables exist only for examined participants (design check).

8.4 Target Analysis

- Class distribution of $y=(DIQ010==1)$ overall (9.8%) and by sex, race, age band, INDFMPIR - identifies subgroups where the event is rarer (affects stratification and metrics).
- Distribution of LBXGH by y - visualize the "controlled diabetics overlap the healthy" zone and the high-A1c undiagnosed group; quantifies label noise of self-report.
- Prevalence vs age curve for DIQ010 - confirms strong age dependence; informs whether age alone is a strong baseline (it will be).
- Co-occurrence matrix among MCQ160B/C/E/F and DIQ010 - the cardiovascular-comorbidity cluster; shows potential secondary targets and why "any CVD" composite may be easier to model.
- Compare self-report (DIQ010) vs lab evidence (LBXGH ≥ 6.5): confusion-style table - reveals undiagnosed vs controlled-diabetic counts, a key insight for framing a screening model.

9. Feature Engineering (from existing columns only)

Source column(s)	New feature	Why useful	Downside	Leakage risk	When to build
BPXOSY1-3	Mean systolic BP (and SD)	Averages reduce measurement noise; SD flags measurement instability.	Loses reading-order info (rarely useful).	None (row-internal).	Before split safe; pipeline for reproducibility.
BPXODI1-3	Mean diastolic BP	Same as above.	None.	None.	Before split safe.
BPXOSY1-3, BPXODI1-3	Pulse pressure = meanSBP - meanDBP	Independent vascular-risk signal beyond SBP/DBP.	Correlated with both inputs.	None.	Before split safe.
BMXWAIST, BMXHIP	Waist-to-hip ratio	Central adiposity measure, stronger than BMI for some outcomes.	Hip circumference is noisy in the obese.	None.	Before split safe.
BMXWT, BMXHT	BMI re-check (BMXBMI exists)	Validate consistency; catch unit/entry errors.	None - BMXBMI already provided.	None.	Validation only.
BMXBMI	Adult BMI category (underweight/normal/overweight/obese I/II/III)	Monotone risk relationship; helps trees and linear models.	Coarser than raw BMI; keep raw too.	None.	Before split safe.
SMQ020, SMQ040, SMD641, SMD030, SMQ050Q/U	Smoking status (never/former/current) + cigarettes-per-day (for current smokers)	Standard public-health composite; much stronger than any single item.	Requires careful skip-logic handling (former-smoker fields missing for never-smokers).	None if built only from these columns.	Before split safe.
PAQ610/615, PAQ625/630, PAQ655/660, PAQ670/675, PAD680	Weekly vigorous+moderate activity minutes; sedentary minutes	Dose-response risk features.	Zero-inflation; reliability of self-reported minutes is modest.	None.	Before split safe.
RIDAGEYR	Age bands (18-39/40-59/60-79/80+) or spline basis	Captures non-linear risk ramp.	Discretization loses info.	None.	Before split safe.
RIDAGEMN	Is-infant flag / exact age for <2	Age precision for children if children kept.	Only 1,033 non-null rows.	None.	Before split safe.
INDFMPPIR	Poverty-ratio bands	Robust to outliers; interpretable.	Modest information loss.	None.	Before split safe.
LBXTC, LBDHDD	Non-HDL cholesterol = TC - HDL	Single atherogenic-lipid summary.	Correlated with components.	None.	Before split safe.
BPAOARM (after decoding)	Arm indicator binary	Tiny but legitimate feature.	Very low variance.	None.	Before split safe.
Missingness indicators	Indicator per important lab/questionnaire block	Structural missingness (fasting, diabetic-follow-up, adult-only) is informative - but see leakage warnings for the diabetic block.	High-dimensionality if overused.	LEAKAGE RISK for any indicator derived from the diabetic-follow-up columns (non-null = diabetic). Use only for non-target-related blocks (fasting, exam attendance).	Before split safe.

10. Machine Learning Problem Formulation

Primary formulation (recommended):

- **X:** demographics (RIAGENDR, RIDAGEYR, RIDRETH3, DMDEDUC2, DMDMARTZ, INDFMPPIR, DMDBORN4), body measures (BMXWT, BMXHT, BMXBMI, BMXWAIST, BMXHIP, BMXARMC, BMXLEG), averaged BP (from BPXOSY/DI1-3), pulse, physical activity summaries, smoking composite, comorbidity flags (MCQ010, MCQ053, MCQ160A, MCQ160D, MCQ300B/C, DIQ080), self-reported high BP (BPQ020) and cholesterol (BPQ080), and HbA1c (LBXGH) if a lab-based screening model is intended.
- **y:** (DIQ010 == 1) binary; drop codes 3/9/NaN -> 14,694 rows, 9.8% positive.
- **Problem type:** binary classification with class imbalance.
- **Baseline:** DummyClassifier (stratified / most-frequent) and a logistic regression on age + BMI + BP - the realistic bar any model must beat.
- **Suitable algorithms:** regularized Logistic Regression (interpretable benchmark), Random Forest / Extra Trees (non-linear benchmark + importance), Histogram Gradient Boosting or XGBoost/LightGBM (typically strongest on tabular), and linear SVM with class weights as an alternative. Tree ensembles tolerate the mixed types after encoding; boosting usually wins on skewed tabular data.

Secondary formulation: regression of LBXGH (HbA1c %) with X as above minus LBXGH; n = 9,737. Use RMSE and R2; MAE is robust to the right tail; report residual plots by subgroup.

Evaluation metrics - and why:

Metric	Problem	Why appropriate here
ROC-AUC	Both	Ranking quality of predicted risk; standard for imbalanced screening; insensitive to the 9.8% threshold choice.
PR-AUC (average precision)	Both	More informative than ROC under 9.8% prevalence; compare against the 0.098 baseline.
Recall (sensitivity)	Classification	In a screening use case, missing a diabetic is costlier than a false alarm.
Precision / F1	Classification	Trade-off summary once an operating threshold is chosen; F1 harmonizes the two in one number.
Macro-F1	Multiclass variant (e.g., 3-class DIQ010 incl. borderline)	Equal weight to minority classes; only if you keep the borderline class.
Confusion Matrix	Classification	Absolute counts at the clinical threshold; required for communicating sensitivity/specificity.
MAE / RMSE / R2	Regression	RMSE penalizes large misses on the high-A1c tail (clinically the worst errors); MAE is the robust companion; R2 shows improvement over the mean predictor.

Accuracy is deliberately NOT recommended: a 90.2% accuracy model that predicts "no diabetes" for everyone is useless, and 90/10 imbalance makes raw accuracy misleading. If you report it, always pair it with the confusion matrix and PR-AUC.

11. Baseline Models (recommendations only - do not train yet)

Model	Purpose in this project	When to use
DummyClassifier (most frequent / stratified)	Reality check on the 9.8% imbalance; every real model must beat this decisively on PR-AUC/F1, not accuracy.	First model to run.
Logistic Regression (L2, class_weight="balanced")	Interpretable linear benchmark; coefficients double as an EDA artifact (which direction do age, BMI, BP point?). Coefficients are in standard-deviation units after scaling.	Reference point and sanity check for feature signs.
Decision Tree (max_depth ~4-6)	A single shallow tree is an inspectable rule benchmark and reveals interaction shapes (e.g., age>60 & BMI>35).	Interpretability baseline.
Random Forest (500 trees, class_weight="balanced_subsample")	Strong non-linear default for mixed tabular data; native feature importance; robust to monotone feature engineering.	Primary tree benchmark.
Histogram Gradient Boosting / XGBoost / LightGBM	Best-in-class for imbalanced tabular data; handles NaN natively (HGB); scale_pos_weight or class weights address imbalance.	Likely final-model family; tune last.
KNN (after scaling)	Distance-based check; rarely wins here but exposes feature-scaling problems early.	Optional.
Naive Bayes / SVM	NB gives a fast probabilistic benchmark; linear SVM with class weights is a good alternative to logistic regression.	Optional.

12. Preprocessing Recommendations

- **1. Drop first:** Unnamed: 0; SDDSRVYR (constant); BMIxxx comment/status columns; MCQ510A-F (near-empty); MCQ230C/D; SI-unit duplicates; WTSAPFRP_P_TRIGLY; administrative flags (SIALANG..., FIA..., MIA..., AIALANGA, SMAQUEX2); and - if DIQ010 is the target - the entire diabetic-follow-up block (DIQ050, DIQ070, DIQ230-360, DID040-350). Keep SEQN as the row index only.
- **2. Fix artifact values:** replace 5.397605346934028e-79 with NaN globally; decode BPAOARM bytes; map NHANES sentinels (7, 9, 99, 7777, 9999 as applicable per codebook) to NaN; map DIQ010 codes 3/9 out of the label.
- **3. Cohort restriction:** for the primary diabetes task restrict to adults (RIDAGEYR >= 18 after fixing epsilon) - this removes a huge structural-missingness block and yields ~9,700+ candidate rows.
- **4. Missing values:** after cohort restriction, use median imputation for numeric features inside a Pipeline; add missingness indicators only for non-leaky blocks (fasting labs, exam attendance). For categorical features, impute the mode or add "Missing" as a level. Never impute the diabetic follow-ups - they are dropped instead.
- **5. Encoding:** binary NHANES items need no one-hot - keep as 0/1 after recoding (1->1, 2->0); 3+-level categories (RIDRETH3, DMDEDUC2, DMDMARTZ, RIDRETH-based, BMDBMIC) via one-hot for linear models (drop_first) and native handling for tree ensembles. Ordinal ones (education, health status HSQ590) may be kept ordinal for trees, one-hot for linear models.
- **6. Scaling:** standardize continuous features (age, BMI, circumferences, BP means, labs, INDFMPIR) for any linear/SVM/KNN/NN model; unnecessary for tree ensembles - so use scaling inside a pipeline branch that only linear models see.
- **7. Outliers:** verify, do not auto-delete: after sentinel removal, remaining extreme values (BMI up to 92.3) are usually real. If a distance-based model struggles, use robust scaling or quantile-based winsorizing fitted on train only.
- **8. Feature selection:** start with the curated list in Section 10; use permutation importance or model-based importance on the tree baseline rather than univariate filters, which would miss redundant-but-strong groups. The SI duplicates must go regardless.
- **9. Dimensionality reduction:** not required for the curated feature set (<40 features). PCA would mostly mix interpretable clinical quantities - avoid for the primary model.
- **10. Imbalance handling:** class_weight in the estimator is the first choice; consider threshold tuning on validation recall/precision rather than resampling; if using SMOTE, apply inside CV folds only.
- **11. Survey weights:** not needed for a predictive ML model, but if you also report population prevalence or associations, use WTINTPRP/WTMECPRP with SDMVPSU/SDMVSTRA (via samplings/statsmodels survey designs). Never feed weights into X.

13. Important Warnings Before Modeling

[!] **TARGET LEAKAGE:** the DIQ diabetic-follow-up block (insulin, pills, specialist visits, self-monitoring) is non-null only for diabetics. Including any of it makes the diabetes task trivial and the model clinically useless.

[!] **LABEL PROXIES:** DIQ160 ("told you are at risk") and the MCD180x "age at diagnosis" variables leak their parent diagnosis targets. Exclude when those diagnoses are targets.

[!] **ID / DESIGN COLUMNS:** SEQN and Unnamed: 0 are pure identifiers; WT*/SDMV* columns encode the survey design - none belong in X.

[!] **EPSILON ARTIFACT:** the 5.397605346934028e-79 values will silently corrupt every statistic if not converted to NaN first.

[!] **BYTES COLUMN:** BPAOARM is stored as b'R'/b'L'/b" and will crash or mislead naive encoders.

[!] **STRUCTURAL MISSINGNESS, NOT MCAR:** 64 columns are >90% missing by design (adult-only, diabetic-only, fasting-only). Never impute across these gates; restrict the cohort instead.

[!] **CHILD/ADULT MIXING:** with 5,867 minors in the file, an all-ages model confounds age with skip-logic missingness. Decide the cohort up front.

[!] **CLASS IMBALANCE:** 9.8% positives makes accuracy misleading; commit to PR-AUC/recall/threshold analysis before tuning.

[!] LABEL NOISE: self-reported diabetes (DIQ010) misses undiagnosed cases and includes well-controlled diabetics with normal labs; quantify the overlap with LBXGH during EDA and frame the use case accordingly.

[!] SENTINEL CODES: 6666/7777/9999-style fills sit inside numeric BP and other columns; a single one would dominate any scaled distance.

[!] FASTING SUBSAMPLE: glucose, triglycerides and LDL exist only for fasted participants - a different (smaller, selected) population; do not treat their absence as ordinary missingness in an all-participant model.

[!] UNWEIGHTED INFERENCE: predictive ML is fine unweighted, but any claim about U.S. population prevalence must use survey weights - do not present unweighted model-derived prevalences as population estimates.

[!] OVERFITTING RISK: ~210 raw columns vs. a ~14k-row labeled set (fewer after cleaning) invites overfitting; enforce the curated feature list, CV-based tuning and a final untouched test set.

14. Recommended Project Roadmap

1. Understand the dataset (this report) - domain, skip logic, survey design.
2. Define the ML problem precisely: cohort (adults 18+), target (DIQ010 binary; secondary LBXGH regression), use case (screening vs risk stratification).
3. Clean the data: drop Unnamed index; epsilon->NaN; decode BPAOARM; sentinel codes->NaN.
4. Remove problematic columns: constants, near-empty, SI duplicates, duplicated weight, administrative flags, and ALL leakage columns (diabetic follow-ups if DIQ010 is the target).
5. Handle missing values: cohort restriction first, then pipeline-fitted median/mode imputation; missingness indicators only for non-leaky blocks.
6. Analyze distributions: histograms/KDEs for numeric variables, counts for categoricals, missingness matrix by prefix.
7. Split the data correctly: stratified train/test after cohort restriction; preprocessing fitted on train only, inside a Pipeline.
8. Perform EDA per Section 8: univariate, bivariate, multivariate, target analysis; log every decision.
9. Feature engineering per Section 9: BP averages, pulse pressure, waist-hip ratio, smoking composite, activity minutes, age bands.
10. Build baselines: Dummy -> Logistic Regression -> Decision Tree -> Random Forest -> Gradient Boosting, all in pipelines with CV.
11. Evaluate with the right metrics: ROC-AUC, PR-AUC, recall, precision, F1, confusion matrix (regression: RMSE/MAE/R²); never accuracy alone.
12. Tune hyperparameters with randomized/grid search inside stratified CV; track runs.
13. Compare models on the held-out test set; check calibration if probabilities will be used clinically.
14. Perform error analysis: examine false negatives (missed diabetics) by age/sex/race; inspect subgroup performance for fairness.
15. Select the final model: simplest model within one standard error of the best performer; document preprocessing and features.
16. Save/deploy if appropriate: joblib pipeline artifact, FastAPI service (fits your learning plan), input validation that rejects epsilon/sentinel values, monitoring for drift.

15. Final Key Takeaways

Most important things to know

- This is NHANES 2017-March 2020 pre-pandemic data (CDC/NCHS): 15,560 participants x 210 merged columns - demographics, body measures, BP, physical activity, smoking, diabetes and medical-condition questionnaires, plus glycohemoglobin, lipid and glucose labs.
- One row = one survey participant; SEQN is a unique ID; there are no duplicate rows.
- The single most valuable modeling target is DIQ010 (doctor-diagnosed diabetes, 9.8% positive after cleaning); the strongest secondary target is LBXGH (HbA1c %, regression).
- The most dangerous trap in the whole file: the diabetic-follow-up questionnaire block leaks the diabetes target perfectly.
- The 5.397605346934028e-79 values are SAS-missing artifacts from the original XPT files - convert to NaN before anything else.
- 64 columns are >90% missing by NHANES skip logic (age gates, fasting subsample, diabetic-only questions) - this is structural, not random.
- 5,867 rows are children; decide your cohort (recommend adults 18+) before splitting.
- BPAOARM is a bytes column (b'R'/b'L') and must be decoded.
- SI-unit duplicates (LBDxxxSI), three LDL methods, three BP readings per person, and a duplicated weight column create redundancy - deduplicate deliberately.
- Random stratified split is correct here: no time axis, no repeated groups; fit all preprocessing inside pipelines on the training fold only.

- Unweighted models are fine for prediction; population-level statements require NHANES survey weights (WTMECPRP + SDMVPSU/SDMVSTRA).
- Self-report targets carry label noise - quantify the DIQ010 vs LBXGH overlap before framing results as a screening tool.

Biggest data-quality concerns

Epsilon artifacts; sentinel codes (6666/7777/9999); the bytes column; near-empty columns; child/adult structural missingness; extreme-but-real anthropometric values.

Biggest leakage risks

Diabetic follow-up block (DIQ050/070/230-360, DID040-350); age-at-diagnosis follow-ups (MCD180x); near-target awareness items (DIQ160/180); ID and survey-design columns.

Most important EDA questions

How much does LBXGH overlap between DIQ010 classes (label noise)? How strong is age alone as a predictor? Which adiposity measure (BMI vs waist vs waist-hip) carries the most signal? Are errors evenly distributed across race/sex/income subgroups?

Recommended ML formulation

Binary classification, adults 18+, $y=(DIQ010==1)$, curated ~30-feature set (demographics + anthropometry + averaged BP + activity + smoking + comorbidity flags + HbA1c if lab-based); secondary regression of LBXGH.

Recommended evaluation metrics

PR-AUC (primary), ROC-AUC, recall, precision, F1, confusion matrix; regression: RMSE, MAE, R2. Never accuracy alone.

Recommended baseline models

Dummy (sanity) -> Logistic Regression (interpretable reference) -> Random Forest (tree benchmark) -> Histogram Gradient Boosting / XGBoost (primary challenger).

References

CDC/NCHS. NHANES 2017-March 2020 Pre-Pandemic Data - Demographics (P_DEMO) codebook:

https://wwwn.cdc.gov/Nchs/Nhanes/2017-2018/P_DEMO.htm

CDC/NCHS. NHANES 2017-March 2020 Pre-Pandemic Glycohemoglobin (P_GHB) documentation:

https://wwwn.cdc.gov/Nchs/Nhanes/2017-2018/P_GHB.htm

CDC/NCHS. NHANES Questionnaires, Datasets and Related Documentation index:

<https://wwwn.cdc.gov/nchs/nhanes/Default.aspx>

CDC/NCHS. NHANES Tutorials (survey weights, design variables, combining cycles):

<https://wwwn.cdc.gov/nchs/nhanes/tutorials/default.aspx>

National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). Diabetes diagnosis & A1c thresholds:

<https://www.niddk.nih.gov/health-information/diagnostic-tests/diabetes>

MedlinePlus (U.S. National Library of Medicine). Glycohemoglobin (HbA1c) test:

<https://medlineplus.gov/lab-tests/hemoglobin-a1c-hba1c-test/>

American Heart Association. Understanding blood pressure readings:

<https://www.heart.org/en/health-topics/high-blood-pressure/understanding-blood-pressure-readings>

Distinction of claim types: facts in Sections 1, 2, 5 and 6 marked "observed in the CSV" were computed directly from mergerd_data.csv. NHANES structure, condition definitions and coding conventions are general domain knowledge from the cited official sources. Roles, targets, metrics, model choices and the roadmap are the author's analytical recommendations.